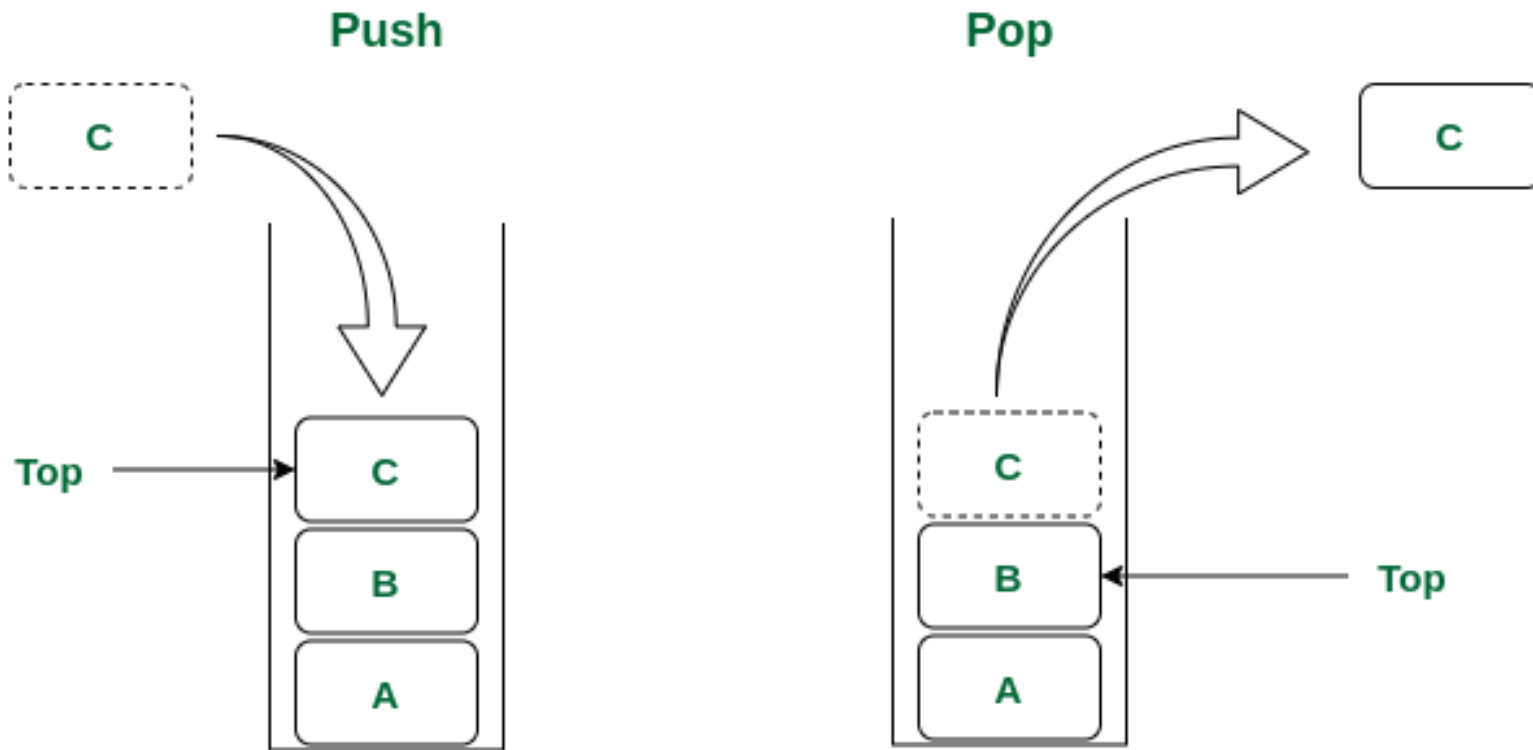




Stack Data Structure

Basic Working of Stack

Stack Follows

LIFO CONCEPT



FILO CONCEPT



FIFO CONCEPT



Major Stack Operations

1. **push():** When we insert an element in a stack then the operation is known as a push. If the stack is full then the overflow condition occurs.
2. **pop():** When we delete an element from the stack, the operation is known as a pop. If the stack is empty means that no element exists in the stack, this state is known as an underflow state.
3. **isEmpty():** It determines whether the stack is empty or not.
4. **isFull():** It determines whether the stack is full or not.'
5. **peek():** Fetch the element from the ith index of the Stack. (i.e. it gives us element from the ith index from the TOP of the stack).
6. **count():** It returns the total number of elements available in a stack.
7. **change():** It changes the element at the ith index from the TOP of the Stack
8. **display():** It prints all the elements available in the stack.

Note:-Remember that - All operations on the stack are performed from the TOP of the Stack.

Key Differences between Algo & Program

Take a Note that Algo is also known as Pseudo Code

Parameter	Algo	Program
TOP	Initialize with value 0	Initialize with value -1
OVERFLOW CONDITION FOR PUSH OPERATION	$TOP \geq N$	$TOP \geq N-1$
UNDERFLOW CONDITION FOR POP OPERATION	$TOP = 0$	$TOP = -1$
UNDERFLOW CONDITION FOR PEEP OPERATION	$TOP - I + 1 \leq 0$	$TOP - I + 1 \leq -1$
UNDERFLOW CONDITION FOR PEEP OPERATION	$TOP - I + 1 \leq 0$	$TOP - I + 1 \leq -1$

Recursion

The recursion means that the method is calling itself again (directly or indirectly). To maintain the previous states, the compiler creates a system stack in which all the previous records of the method are maintained.

Basic Recursion Programs and MCQs

Tower of Hanoi Concept

String Reversal Using Stack

Decimal to Binary Using Stack

TO BE DONE FROM
NOTES OR FOLLOW
THE PROGRAMS BY
YOUR OWN LOGIC
DONE IN CLASS